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Optimal exit from Uniswap v3 and best expected return for a liquidity provider

Ankush AGARWAL* Emmanuel GOBET† ‡ §

November 26, 2025

Abstract

We analyze the profitability of liquidity providers' (LPs) positions in Uniswap v3 by aggregating fee income and impermanent loss within an optimal stopping framework. Our first result shows that the liquidity burn should be optimized over one range at a time, rather than simultaneously. Second, without discounting future fees, there is no finite optimal liquidity burn time and indefinite liquidity provision is optimal. In this case, we derive closed-form expressions for the value of LP positions according to different price levels of liquidity burn. Third, with a discount factor, we introduce an equivalent rate of return and demonstrate that under a Black-Scholes model with volatility σ, the optimal return is approximately $0.425 \cdot \sigma^2$ (i.e. about 10% return for 50% volatility), and it is achieved by choosing the at-the-money range of liquidity. These results provide explicit formulas and strategic insights for LPs in Uniswap v3, and complement recent works on Uniswap v2 and fee modelling by highlighting the distinct impact of concentrated liquidity.

Keywords: automated market maker, concentrated liquidity, optimal stopping problem, best return

1 Introduction

Context. In recent years, decentralized finance (DeFi) has emerged as one of the most disruptive innovations in the financial industry, challenging traditional intermediaries and reshaping how

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markets function. By leveraging blockchain technology and smart contracts, DeFi protocols enable open, transparent, and permissionless access to financial services, ranging from lending and derivatives to asset management, see [GM23] for an overview. Among these innovations, automated market makers (AMMs) represent a cornerstone mechanism for decentralized trading, eliminating the need for centralized order books while ensuring continuous liquidity. AMMs allow users to act simultaneously as liquidity providers and traders, creating a self-sustaining market where prices are algorithmically determined. This model has not only democratized market access but also introduced new forms of financial engineering, incentivization, and risk management.

Uniswap v3. Uniswap, as one of the leading AMM protocols, has undergone several iterations, with Uniswap v3 [AZS+21] introducing concentrated liquidity as a major breakthrough. Namely, the protocol brings together two categories of participants in a given pool[1]: liquidity providers (LPs) and swap traders. LPs allocate pairs of tokens (*liquidity mint*), say X and Y, across predefined price intervals, thereby concentrating their liquidity where they expect trading activity to occur. Swap traders, in turn, interact with this liquidity by exchanging one token for the other (swapping X for Y or vice versa). Through each of these transactions, traders pay a fee that is automatically distributed to the LPs who have supplied liquidity at the relevant price range. However, swap traders do not only generate fees; by executing trades they also alter the relative quantities of tokens held in the pool, thereby shifting the effective price. As a result, when an LP withdraws its position (*liquidity burn*), the composition of tokens received may differ from what would have been obtained under a passive buy-and-hold strategy, commonly referred to as HODL. The discrepancy between these two outcomes is known as impermanent loss [LHRW21], and it represents a central risk faced by liquidity providers. These mechanisms have been thoroughly analyzed and quantified in [EGM25], to which we refer the reader for a comprehensive treatment of Uniswap's design and an extensive set of references. The main outputs of this analysis will be recalled in Section 2.

Statement of the problem and main contributions. Our objective is to analyze the profitability of liquidity providers' positions, aggregating both the risk of impermanent loss and the income generated from fees, with a focus on identifying the optimal exit time from the pool. We formalize this problem within the framework of optimal stopping theory. Our main results can be summarized as follows. First, it is more efficient to optimize range by range of concentrated liquidity rather than to liquidate all ranges simultaneously. Second, in the absence of discounting, we prove that no finite optimal stopping time exists, and that the strategy of remaining in the pool forever is optimal. Within this framework, we are able to derive closed-form expressions for the value of a position when liquidity is burned at a certain price level. Finally, when discounting is introduced, we can define an equivalent rate of return associated with the liquidity position and

[1] According to [Tea25], as of May 2025, there are about almost 30 000 pools on Uniswap v3.

show that, under a Black-Scholes model with volatility σ, this best return is $c^\star\sigma^2$ where $c^\star \approx 0.425$. It is obtained when the LP allocates tokens to the range that includes the current X-Y price at the time liquidity is added.

We would like to mention two recent works [CZ25] and [DFS25], whose topics are closely related to this paper, but which differ quite significantly from our current contributions. In [CZ25], the authors study a similar optimal stopping problem in Uniswap v2. Since their AMM rules differ significantly from those of Uniswap v3, results can not be compared although they also found (in the case of zero discount factor) that the LP may have an interest to stay in the pool forever. The authors of [DFS25] assume that the LP wealth process must be a martingale (by invoking no-arbitrage arguments, which stands for an interesting but debatable hypothesis in that context) and evaluate fees implicitly from the fair price of impermanent loss, which is obtained by model calibration from options data. Our approach is quite different since we consider the real value of fees as presented in [EGM25].

Plan of the paper. Section 2 provides an overview of the state-of-the-art results previously established for Uniswap v3, presented in a form which is directly applicable to our analysis. Section 3 gathers our main new results on optimal stopping problems for general price processes. We specialize our study to the martingale case where a time-change argument allows to reduce to the constant volatility model, and where we show by explicit computations that it is optimal for the LP to stay in the pool forever, irrespective of the liquidity range. Then, in Section 4 we investigate what is the equivalent return for an LP investing in the pool. Some technical results are postponed to the appendix.

2 Overview of Uniswap V3

Consider an X/Y pool where tokens of type X and Y can be exchanged, for example, USDC and ETH. The price p refers to the rate $X - Y$ with the foreign exchange convention, i.e., 1 token of X is worth p tokens of Y. Different notions of price will be introduced in the following, like the pool price and another reference price (e.g., that of a centralized exchange (CEX)).

2.1 Uniswap v3: pool parameters

In Uniswap v3 pools, prices can evolve with arbitrary values, but the fees owed to each liquidity provider (LP) depend on the liquidity made available across different ranges in which the pool price evolves. The bounds of these price ranges are of the form

$$p(\tau) := 1.0001^\tau, \tag{2.1}$$

for some $\tau \in \mathbb{Z}$, taking values in a set of signed integer values called *ticks*. However, not any tick ranges can be used to add liquidity. The ranges over which liquidity can be added are a multiple of a fixed number of ticks δ_π, which is determined at the setting of the pool based on the swap fees ϕ. When a Uniswap v3 pool is set, the possible choices by default[2] are $\delta_\pi = 10, 60, 200$, based on the corresponding swap fees $\phi = 0.05\%, 0.3\%, 1\%$, respectively. Thus, the smallest range $R(i)$ on which liquidity can be added is of the form

$$R(i) := \Big[p(i\delta_\pi), p((i+1)\delta_\pi)\Big), \qquad \text{for } i \in \mathbb{Z},$$

and is called a *unitary range*.
It is common to use the notation π for the square-root of price p. We write

$$\pi(i) := \sqrt{p(i\delta_\pi)}.$$

Similarly, the square root price range is defined by

$$\rho(i) := \Big[\pi(i\delta_\pi), \pi((i+1)\delta_\pi)\Big) \qquad \text{for } i \in \mathbb{Z}.$$

The liquidity minted by an LP on the unitary range $R(i)$ is denoted ΔL_i. We will frequently use the notion of minted liquidity curve $(\Delta L_\pi)_{\pi>0}$ where for notational convenience we set $\Delta L_\pi = \Delta L_i$ for the unique index i such that $\pi^2 = p \in R(i)$. See the left plot of Figure 1 where an example of total liquidity $(L_\pi)_{\pi>0}$ is depicted.
For the sake of completeness, we recall the relationship between numbers x_r and y_r of *real tokens* of X and Y, respectively, and liquidity L available in a price range $R(i)$:

$$\left(x_r + \frac{L}{\pi(i+1)}\right)(y_r + L\pi(i)) = L^2; \tag{2.2}$$

see [EGM25, Section 2.1] for a rigourous derivation from the Uniswap v3 source code. This formula is known as the CPMM (Constant Product Market Maker) formula for Uniswap v3. An example of this rule is depicted in the right hand side of Figure 1. The quantities $x_r + \frac{L}{\pi(i+1)}$ and $y_r + L\pi(i))$ are referred to as *virtual quantities* of tokens.

2.2 Swap fees

We aim to analyse the situation of an LP who wants to benefit from future swap trades fees by providing liquidity in the tokens X and Y. As shown in [EGM25, Theorem 3.3], for a given minted liquidity curve $(\Delta L_\pi)_\pi$, at time t the value of an LP's position net of swap fees in terms of Y token is given as

$$V_P(t) = p_t \int_{\pi_t}^{\infty} \frac{\Delta L_\pi}{\pi^2} \mathrm{d}\pi + \int_0^{\pi_t} \Delta L_\pi \mathrm{d}\pi, \tag{2.3}$$

[2] although it is possible to have pools with other tick spacings and fees.

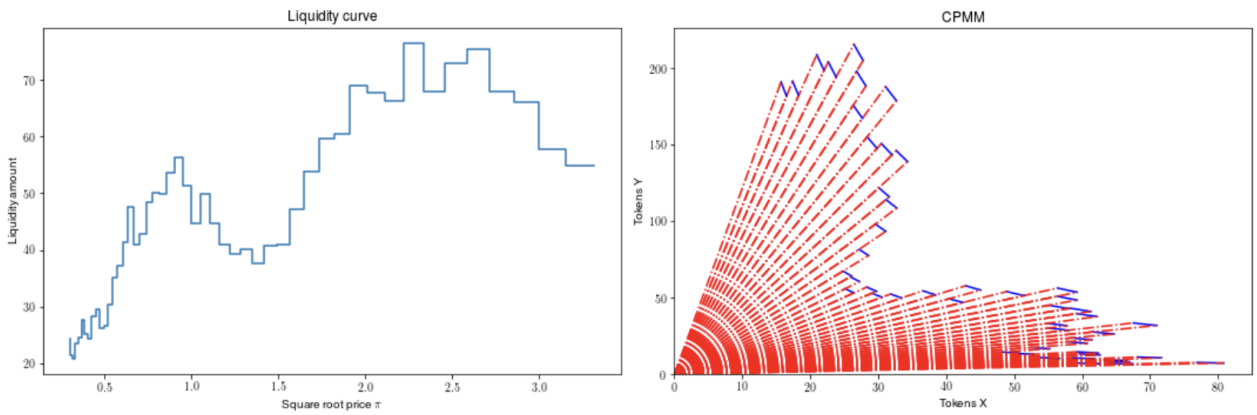

Figure 1: Different liquidity quantities L_π deposited across several consecutive square root price ranges (on the left). On the right, the representation of the CPMM formula (2.2) for each range as a function of virtual tokens, using the liquidity specific to the range.

where p_t is the pool price at time t and $\pi_t = \sqrt{p_t}$ is the square root price. The above equation holds in full generality and results from the real mechanism of Uniswap v3 AMM. It is shown in [EGM25, Corollary 3.4.] that the second order derivative of $V_P(t)$ with respect to p_t is negative, yielding a loss when compared with a static strategy (called HODL).
At the same time, the amounts of fees earned from swap trading over a period of time by the LP that has provided the liquidity curve $(\Delta L_\pi)_\pi$ can be approximated in closed-form, see [EGM25, Theorem 4.1]. The working assumptions are as follows.

(H1) The pool price p evolves at discrete times, and at those discrete times, it coincides with a latent price modelled by a general continuous-time Itô process

$$\frac{\mathrm{d}p_t}{p_t} = \mu_t \mathrm{d}t + \sigma_t \mathrm{d}W_t, \tag{2.4}$$

defined on on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0\leq t\leq T}, \mathbb{P})$ that supports a Brownian motion W, and with a filtration that satisfies the "usual conditions" (right-continuity, completeness). Here, we assume that the drift $(\mu_t)_t$ and volatility $(\sigma_t)_t$ coefficients are progressively-measurable and bounded, furthermore the volatility σ is bounded away from 0.

(H2) In addition, the swap trades occur when the latent price process $(p_t)_t$ changes by $\phi\%$, i.e the swap times $(t_i)_i$ are described by

$$t_{i+1} := \inf\left\{t \geq t_i : p_t = p_{t_i}(1+\phi)^{\pm 1}\right\},$$

with the convention that $t_0 := 0$.

Assumption (H2) is coherent with the arbitrage model of Angeris et al. [AKC+21, Section 2.1], where a trade occurs when the pool price differs by $\phi\%$ from a reference price, so that after the swap trade, both prices equal.
Under those assumptions, the authors of [EGM25, Theorem 4.1] prove that the numbers of X and Y tokens collected on the period $[0,t]$ as swap fees by the LP converge in probability, as $\phi \to 0$, to Φ_t^X and Φ_t^Y respectively, where

$$\Phi_t^X := \int_0^\infty \Delta L_{b^{1/2}} \frac{A_t^b(p)}{4b^{5/2}} \mathrm{d}b, \qquad \Phi_t^Y := \int_0^\infty \Delta L_{b^{1/2}} \frac{A_t^b(p)}{4b^{3/2}} \mathrm{d}b, \tag{2.5}$$

and where $A_t^b(p)$ is the local time at time t and level b of the process p, i.e.

$$A_t^b(p) := 2\left[(p_t - b)_+ - (p_0 - b)_+ - \int_0^t \mathbf{1}_{p_s > b} \mathrm{d}p_s\right].$$

See [RY99, Theorem 1.2, p.222].

2.3 Problem statement for a full liquidity curve

We now study a possible optimal exit or optimal liquidity burn time for an LP for which the objective function at time t is the sum of the value of the pool liquidity (2.3) and the swap trade fees in terms of token Y (2.5):

$$J\big(t, (\Delta L_\pi)_\pi\big) = p_t \int_{\pi_t}^\infty \frac{\Delta L_\pi}{\pi^2} \mathrm{d}\pi + \int_0^{\pi_t} \Delta L_\pi \mathrm{d}\pi + p_t \Phi_t^X + \Phi_t^Y. \tag{2.6}$$

Then, given a liquidity curve $(\Delta L_\pi)_\pi$, the optimal stopping problem is expressed as

$$\sup_{t \in \mathcal{T}(\mathbb{R}^+)} \mathbb{E}\left[J\big(t, (\Delta L_\pi)_\pi\big)\right], \tag{2.7}$$

where the set $\mathcal{T}(A)$ (for $A \subset [0, +\infty)$) stands for the set of $\mathcal{F}$-stopping times taking values in A with probability 1. Since $J\big(t, (\Delta L_\pi)_\pi\big)$ is non-negative, without extra integrability condition, the above expectation may be equal to $+\infty$. Note that stopping times taking the value $+\infty$ with positive probability 1 are not in $\mathcal{T}(\mathbb{R}^+)$.

We compare the functional cost (2.7) when the LP burns the entire liquidity curve $(\Delta L_\pi)_\pi$ at a single stopping time, compared to the situation where it burns each piece of liquidity ΔL_π at different stopping times τ_π. Our first result is the following.

Proposition 2.1. *It is systematically better for the LP to burn each piece of liquidity ΔL_π at different stopping times τ_π than to burn the entire liquidity curve $(\Delta L_\pi)_\pi$ at once.*

The proof is given in Section A.1.

2.4 Problem statement for a single liquidity range

From Proposition 2.1, it is clear that there is no interest for an LP to liquidate its entire position at once. Therefore, we focus only on the case of burning liquidity on a single *unitary* price range $R = [p_\ell, p_u)$ (associated to the square root price range $\rho = [\pi_\ell, \pi_u)$), with $\Delta L_{\pi_\ell} = 1$, and consider the following optimal stopping problem:

$$\sup_{t \in \mathcal{T}(\mathbb{R}^+)} \mathbb{E}\left[J_R(t)\right], \tag{2.8}$$

where $J_R(.)$ is given by

$$J_R(t) := p_t \int_{\pi_t}^{\infty} \mathbf{1}_{\pi \in \rho} \frac{1}{\pi^2} \mathrm{d}\pi + \int_0^{\pi_t} \mathbf{1}_{\pi \in \rho} \mathrm{d}\pi + p_t \int_R \frac{A_t^b(p)}{4b^{5/2}} \mathrm{d}b + \int_R \frac{A_t^b(p)}{4b^{3/2}} \mathrm{d}b.$$

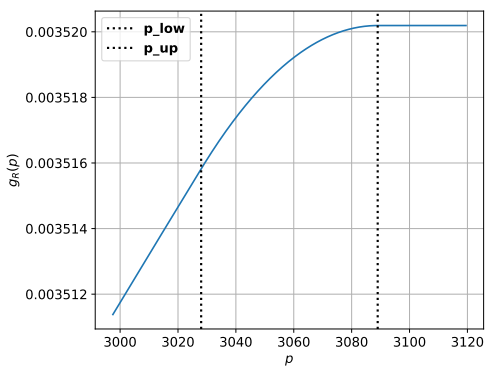

Figure 2: Graph of $g_R(.)$ for $R := [p_\ell, p_u)$ with $p_\ell = 3028$ and $p_u = 3089$.

Set

$$g_R(p_t) := p_t \int_{\pi_t}^{\infty} \mathbf{1}_{\pi \in \rho} \frac{1}{\pi^2} \mathrm{d}\pi + \int_0^{\pi_t} \mathbf{1}_{\pi \in \rho} \mathrm{d}\pi.$$

A few straightforward computations show that

$$g_R(p_t) = \left(\frac{1}{\pi_t^R} - \frac{1}{\pi_u}\right) p_t + \left(\pi_t^R - \pi_\ell\right),$$

where π^R is the projection of π on the range R:

$$\pi^R = \begin{cases} \pi_u \text{ if } \pi_u < \pi, \\ \pi_\ell \text{ if } \pi < \pi_\ell, \\ \pi \text{ otherwise.} \end{cases}$$

Besides, $g_R(.)$ is concave and non-negative, see Figure 2.

3 Optimal stopping problem in the submartingale case

In this section, we focus on the case when p (2.4) is a submartingale. This includes the usual case of a martingale when $\mathbb{P}$ is a risk-neutral measure used for valuation of options written on p, and also the case of non-negative drift for p, which we have often observed in past years with a bullish market.

(Hsub) p is a submartingale, i.e., $\mu_s \geq 0$, for any s almost surely.

3.1 A first simplification of the reward $J_R(.)$

We use the submartingale assumption (Hsub) to derive a simplification of the optimal stopping problem (2.8). To see this, we use the occupation-time formula (see [RY99, Corollary 1.6, p.224]) and transform the local time integral into a time integral. We then get

$$J_R(t) = g_R(p_t) + \frac{1}{4} p_t \mathcal{I}_t^{-\frac{1}{2}} + \frac{1}{4} \mathcal{I}_t^{+\frac{1}{2}}, \tag{3.1}$$

where for any $\alpha \in \mathbb{R}$, we have set

$$\mathcal{I}_t^{\alpha} := \int_0^t \sigma_s^2 p_s^{\alpha} \mathbf{1}_{p_s \in R} \mathrm{d}s.$$

Lemma 3.1. *Under **(H1)**, **(H2)** and **(Hsub)**, for any stopping time $t \in \mathcal{T}(\mathbb{R}^+)$, we have*

$$\mathbb{E}\left[p_t \mathcal{I}_t^{-\frac{1}{2}}\right] \geq \mathbb{E}\left[\mathcal{I}_t^{+\frac{1}{2}}\right],$$

where the inequality becomes an equality in the martingale case $\mu \equiv 0$.

Proof. All quantities are non-negative and we can extensively use the Fubini theorem, in order to write

$$\begin{aligned}
\mathbb{E}\left[p_t \mathcal{I}_t^{-\frac{1}{2}}\right] &= \int_0^{+\infty} \mathbb{E}\left[\mathbf{1}_{s<t} \sigma_s^2 p_t p_s^{-\frac{1}{2}} \mathbf{1}_{p_s \in R}\right] \mathrm{d}s \\
&= \int_0^{+\infty} \mathbb{E}\left[\mathbf{1}_{s<t} \sigma_s^2 \mathbb{E}\left[p_t \mid \mathcal{F}_s\right] p_s^{-\frac{1}{2}} \mathbf{1}_{p_s \in R}\right] \mathrm{d}s \\
&\geq \int_0^{+\infty} \mathbb{E}\left[\mathbf{1}_{s<t} \sigma_s^2 p_s^{+\frac{1}{2}} \mathbf{1}_{p_s \in R}\right] \mathrm{d}s = \mathbb{E}\left[\mathcal{I}_t^{+\frac{1}{2}}\right].
\end{aligned}$$

The inequality follows from the submartingale assumption (Hsub). □

In other words, in the martingale case, the expected values of fees collected in X and Y coincide, which is intuitively coherent. Therefore, combining the above with (3.1) leads directly to the following result.

Corollary 3.2. *Under **(H1)**, **(H2)** and **(Hsub)**, the optimal stopping problem (2.8) is bounded from below as follows:*

$$\sup_{t\in\mathcal{T}(\mathbb{R}^+)} \mathbb{E}\left[J_R(t)\right] \geq \sup_{t\in\mathcal{T}(\mathbb{R}^+)} \mathbb{E}\left[J_R^{sub}(t)\right] \qquad \text{with} \qquad J_R^{sub}(t) = g_R(p_t) + \frac{1}{2}\mathcal{I}_t^{+\frac{1}{2}},$$

where the inequality becomes an equality in the martingale case $\mu \equiv 0$.

3.2 Theoretical properties of the optimal stopping problem on J_R^{sub}

This optimal stopping problem has an explicit solution owing to the submartingale property of the reward process.

Theorem 3.3. *Under **(H1)**, **(H2)** and **(Hsub)**, the process $(J_R^{sub}(t) : t \geq 0)$ is a non-negative submartingale and*

$$\sup_{t\in\mathcal{T}(\mathbb{R}^+)} \mathbb{E}\left[J_R^{sub}(t)\right] = \lim_{t\uparrow+\infty} \mathbb{E}\left[J_R^{sub}(t)\right] \geq \frac{1}{2}\mathbb{E}\left[\int_0^{+\infty} \sigma_s^2 p_s^{1/2} \mathbf{1}_{p_s\in R} \mathrm{d}s\right]. \tag{3.2}$$

The inequality is an equality in the martingale case $\mu \equiv 0$.

Proof. A direct computation (see [EGM25, Corollary 3.5]) shows that $g_R(.)$ is a piecewise C^2-function and that

$$g_R'(p) = \int_{\sqrt{p}}^{+\infty} \frac{\mathbf{1}_{\pi^2\in R}}{\pi^2}\mathrm{d}\pi, \qquad g_R''(p) = -\frac{\mathbf{1}_{p\in R}}{2p^{3/2}},$$

for almost every p. Recall that based on our definitions, the notations $\pi^2 \in R$, $\pi \in \rho$, and $p \in R$ are equivalent. Hence, the Itô formula for piecewise C^2-functions gives

$$\begin{aligned}\mathrm{d}J_R^{sub}(t) &= g_R'(p_t)\mathrm{d}p_t + \frac{1}{2}g_R''(p_t)\sigma_t^2 p_t^2 \mathrm{d}t + \frac{1}{2}\sigma_t^2 p_t^{1/2}\mathbf{1}_{p_t\in R}\mathrm{d}t \\ &= g_R'(p_t)p_t\big(\mu_t \mathrm{d}t + \sigma_t \mathrm{d}W_t\big) + \frac{1}{4}\sigma_t^2 p_t^{1/2}\mathbf{1}_{p_t\in R}\mathrm{d}t.\end{aligned}$$

Since $g_R'(p_t)$ and μ_t are non-negative, $J_R^{sub}(t)$ is a submartingale. The first equality in (3.2) then directly follows from the optional sampling theorem, while the inequality comes from ignoring the non-negative term $\lim_{t\uparrow+\infty} \mathbb{E}\left[g_R(p_t)\right]$. In the martingale case $\mu \equiv 0$. Since p can be seen as a time-changed geometric Brownian motion, the limits of p_t and $\mathbb{E}\left[g_R(p_t)\right]$ as $t \to +\infty$ are 0. □

Thus, in the submartingale case, an LP is better off to stay in the pool forever. It shows that collecting fees really incentivize actors to provide liquidity in a Uniswap v3 pool. This is a very significative result in practice. It is clear from the proof that the accuracy of the inequality sign depends on $\lim_{t\uparrow+\infty} \mathbb{E}\left[g_R(p_t)\right]$. This limit exists since the expectation is essentially the difference

of two convergent quantities. Alternatively, if we had known or assumed the limit of p at $+\infty$, whenever it exists, we could have deduced the limit of $\mathbb{E}\left[g_R(p_t)\right]$ by an application of the bounded convergence theorem. In the martingale case $\mu \equiv 0$ and since p can be seen as a time-changed geometric Brownian motion, the limits of p_t and $\mathbb{E}\left[g_R(p_t)\right]$ are 0. Consequently, $\lim_{t\uparrow+\infty}\mathbb{E}\left[J_R^{sub}(t)\right]$ is a good candidate for the lower bound of $\sup_{t\in\mathcal{T}(\mathbb{R}^+)}\mathbb{E}\left[J_R(t)\right]$ from Corollary 3.2.
Next, we prove that the lower-bound in (3.2), in the martingale case, does not depend on the volatility. It means that in the martingale case, $\sup_{t\in\mathcal{T}(\mathbb{R}^+)}\mathbb{E}\left[J_R(t)\right]$ results in an explicit formula.

Theorem 3.4. *Under assumptions **(H1)**, **(H2)** with $\mu = 0$, the lower bound*

$$\mathbb{E}\left[\int_0^{+\infty}\sigma_s^2 p_s^{1/2}\mathbf{1}_{p_s\in R}\mathrm{d}s\right]$$

in (3.2) is finite and does not depend on the volatility parameter σ.

Proof of Theorem 3.4. Let us first prove that the expectation is finite. Observe that the trivial bound $\int_0^t \sigma_s^2 p_s^{1/2}\mathbf{1}_{p_s\in R}\mathrm{d}s \le t|\sigma|_\infty^2\pi_u$ does not lead to the required estimate because of the linear dependence in t. Instead, we use the exponential form of p and time-changed Brownian motion. Set $A_t := \int_0^t \sigma_s^2\mathrm{d}s$, and $\beta_u := \int_0^{A_u^{-1}}\sigma_s\mathrm{d}W_s$. A is a continuous and strictly increasing process. By the Dambis-Dubins-Schwarz theorem [RY99, Theorem 1.6, Chapter V.1], β is a Brownian motion and we get

$$p_{A_u^{-1}} = p_0\exp(-\frac{1}{2}\int_0^{A_u^{-1}}\sigma_s^2\mathrm{d}s + \int_0^{A_u^{-1}}\sigma_s\mathrm{d}W_s) = p_0\exp(-\frac{1}{2}u + \beta_u).$$

This technique of time-change is well known in solving optimal stopping problems, see [PS06, Chapter IV, Section 10]. Then

$$\begin{aligned}
0 \le \sup_{t\ge0}\int_0^t \sigma_s^2 p_s^{1/2}\mathbf{1}_{p_s\in R}\mathrm{d}s &= \int_0^{+\infty} p_{A_v^{-1}}^{1/2}\mathbf{1}_{p_{A_v^{-1}}\in R}\mathrm{d}v \\
&\le \int_0^{+\infty}\sqrt{p_0}\exp\left(-\frac{1}{4}v + \frac{1}{2}\beta_v\right)\mathrm{d}v, \text{ then,} \\
\mathbb{E}\left[\sup_{t\ge0}\int_0^t \sigma_s^2 p_s^{1/2}\mathbf{1}_{p_s\in R}\mathrm{d}s\right] &\le \sqrt{p_0}\int_0^{+\infty}\mathbb{E}\left[\exp\left(-\frac{1}{4}v + \frac{1}{2}\beta_v\right)\right]\mathrm{d}v \\
&= \sqrt{p_0}\int_0^{+\infty}\exp\left(-\frac{1}{8}v\right)\mathrm{d}v < +\infty.
\end{aligned}$$

□

3.3 LP position value of the early-exit strategy at the first hitting time of a level

In the previous section, we showed that in the in the submartingale case, an LP is better off to stay in the pool forever. However, some LPs who consider that market dynamics are driven by the crossing of psychological levels may favour strategies which are associated with the first hitting of a price level λ. Thus, in this section we derive the value of these suboptimal strategies. To conduct this analysis, we first restrict to the martingale case $\mu = 0$. Recall from Theorem 3.4 and the Dambis-Dubins-Schwarz time change result, the LP position value will not depend on the volatility process. Then, we can restrict to the case of a geometric Brownian motion with constant volatility $\sigma > 0$. Since the final formulas will be independent of σ, we can even take $\sigma = 1$. However, in Section 4, we consider the situation when swap fess is discounted with a nonzero discount factor. To prepare for that case, we keep the notation with $\sigma > 0$, not necessarily equal to one. Recall from Corollary 3.2, the value of the early-exit strategy when the LP chooses to burn liquidity at the price level λ is given by

$$V^{\lambda}(p_0) := \mathbb{E}\left[J_R^{sub}(p_{t^\lambda})\right], \tag{3.3}$$

where t^λ is the first hitting time of a price level λ by p. It is defined as

$$t^\lambda := \inf\{s : p_s = \lambda\}, \tag{3.4}$$

with the convention that $\inf \emptyset = +\infty$. In the above, the exit level λ can be either below or above the current price level p_0. Since $p_t \to 0$ as $t \to +\infty$, the first hitting time t^λ is finite with probability one if $\lambda \leq p_0$, and infinite with positive probability when $\lambda > p_0$. See (3.9) for the exact probability value.

3.3.1 Preliminary results

To evaluate $V^\lambda(p_0)$, we will take advantage of explicit expressions for expectations of occupation time functionals stopped at t^λ. In [BS15], the authors derived several properties of local time defined with respect to the Lebesgue measure

$$\mathcal{A}_T^b(p) := \lim_{\epsilon \downarrow 0} \frac{1}{\epsilon} \int_0^T \mathbf{1}_{p_t \in [b, b+\epsilon)} \mathrm{d}t. \tag{3.5}$$

In our analysis in previous sections, we have defined the local time $A_\cdot^\cdot(.)$ with respect to the speed measure associated with the martingale price process p_t with constant volatility σ. The two definitions are related in the following way:

$$A_T^b(p) := \lim_{\epsilon \downarrow 0} \frac{1}{\epsilon} \int_0^T \mathbf{1}_{p_t \in [b, b+\epsilon)} \mathrm{d}\langle p \rangle_t = \lim_{\epsilon \downarrow 0} \frac{1}{\epsilon} \int_0^T \mathbf{1}_{p_t \in [b, b+\epsilon)} \sigma^2 b^2 \mathrm{d}t = \sigma^2 b^2 \mathcal{A}_T^b(p). \tag{3.6}$$

Proposition 3.5. *[BS15, Formula 2.3.1 p.623] The Laplace transform of the local time at level b as defined in (3.5) stopped as it hits a level $\lambda \le b$, for geometric Brownian motion $p_t = p_0 \exp\big(-\frac{1}{2}\sigma^2 s + \sigma W_s\big)$ is given as*

$$\begin{aligned}
&\mathbb{E}\left[\mathrm{e}^{-\gamma \mathcal{A}^b_{t^\lambda}(p)} \mathbf{1}_{t^\lambda<\infty} \mid p_0\right] \\
&= \left(\frac{\lambda}{p_0}\right)^{-1/2} \exp\left(-\frac{1}{2}|\log(\lambda/p_0)|\right) \times \\
&\begin{cases} 1 & \text{in case (i): } p_0 \le \lambda \le b, \quad b \le \lambda \le p_0, \\ \frac{\frac{1}{2}b\sigma^2+\gamma(1-\mathrm{e}^{-|\log(p_0/b)|})}{\frac{1}{2}b\sigma^2+\gamma(1-\mathrm{e}^{-|\log(\lambda/b)|})}, & \text{in case (ii): } b\wedge\lambda \le p_0 \le b\vee\lambda, \\ \frac{\frac{1}{2}b\sigma^2}{\frac{1}{2}b\sigma^2+\gamma(1-\mathrm{e}^{-|\log(\lambda/b)|})}, & \text{in case (iii): } \lambda \le b \le p_0, \quad p_0 \le b \le \lambda. \end{cases}
\end{aligned}$$

For $\lambda > b$, it is given as

$$\begin{aligned}
&\mathbb{E}\left[\mathrm{e}^{-\gamma \mathcal{A}^b_{t^\lambda}(p)} \mathbf{1}_{t^\lambda=\infty} \mid p_0\right] \\
&= \begin{cases} 0 & \text{in case (iv): } p_0 \le \lambda \le b, \quad b \le \lambda \le p_0, \\ \frac{\frac{1}{2}b\sigma^2(1-\mathrm{e}^{-|\log(p_0/\lambda)|})}{\frac{1}{2}b\sigma^2+\gamma(1-\mathrm{e}^{-|\log(\lambda/b)|})}, & \text{in case (v): } b\wedge\lambda \le p_0 \le b\vee\lambda, \\ (1-\mathrm{e}^{-|\log(p_0/b)|}) + \frac{\frac{1}{2}b\sigma^2(\mathrm{e}^{-|\log(p_0/b)|}-\mathrm{e}^{-|\log(p_0/\lambda)|})}{\frac{1}{2}b\sigma^2+\gamma(1-\mathrm{e}^{-|\log(\lambda/b)|})}, & \text{in case (vi): } \lambda \le b \le p_0, \quad p_0 \le b \le \lambda. \end{cases}
\end{aligned}$$

We can obtain similar expressions for $A^b_T(p)$ defined in (3.6) by using basic algebraic manipulation in Proposition 3.5.

Proposition 3.6. *The Laplace transform of the local time at level b as defined in (3.6) stopped as it hits a level $\lambda \le b$, for geometric Brownian motion $p_t = p_0 \exp\big(-\frac{1}{2}\sigma^2 s + \sigma W_s\big)$ is given as*

$$\begin{aligned}
&\mathbb{E}\left[\mathrm{e}^{-\gamma A^b_{t^\lambda}(p)} \mathbf{1}_{t^\lambda<\infty} \mid p_0\right] \qquad (3.7) \\
&= \left(\frac{\lambda}{p_0}\right)^{-1/2} \exp\left(-\frac{1}{2}|\log(\lambda/p_0)|\right) \times \\
&\begin{cases} 1 & \text{in case (i): } p_0 \le \lambda \le b, \quad b \le \lambda \le p_0, \\ \frac{\frac{1}{2}b\sigma^2+\gamma b^2\sigma^2(1-\mathrm{e}^{-|\log(p_0/b)|})}{\frac{1}{2}b\sigma^2+\gamma b^2\sigma^2(1-\mathrm{e}^{-|\log(\lambda/b)|})}, & \text{in case (ii): } b\wedge\lambda \le p_0 \le b\vee\lambda, \\ \frac{\frac{1}{2}b\sigma^2}{\frac{1}{2}b\sigma^2+\gamma b^2\sigma^2(1-\mathrm{e}^{-|\log(\lambda/b)|})}, & \text{in case (iii): } \lambda \le b \le p_0, \quad p_0 \le b \le \lambda. \end{cases}
\end{aligned}$$

For $\lambda > b$, it is given as

$$\mathbb{E}\left[\mathrm{e}^{-\gamma A^b_{t^\lambda}(p)} \mathbf{1}_{t^\lambda=\infty} \mid p_0\right] \qquad (3.8)$$

$$= \begin{cases} 0 & \textit{in case (iv): } p_0 \le \lambda \le b, \quad b \le \lambda \le p_0, \\ \frac{\frac{1}{2}b\sigma^2(1-\mathrm{e}^{-|\log(p_0/\lambda)|})}{\frac{1}{2}b\sigma^2+\gamma b^2\sigma^2(1-\mathrm{e}^{-|\log(\lambda/b)|})}, & \textit{in case (v): } b \wedge \lambda \le p_0 \le b \vee \lambda, \\ (1-\mathrm{e}^{-|\log(p_0/b)|}) + \frac{\frac{1}{2}b\sigma^2(\mathrm{e}^{-|\log(p_0/b)|}-\mathrm{e}^{-|\log(p_0/\lambda)|})}{\frac{1}{2}b\sigma^2+\gamma b^2\sigma^2(1-\mathrm{e}^{-|\log(\lambda/b)|})}, & \textit{in case (vi): } \lambda \le b \le p_0, \quad p_0 \le b \le \lambda. \end{cases}$$

Moreover, from equation (3.7), with $\gamma = 0$, we get the following result.

Corollary 3.7. *The hitting time t^λ defined in (3.4) for geometric Brownian motion $p_t = p_0 \exp\big(-\frac{1}{2}\sigma^2 s + \sigma W_s\big)$ satisfies the following condition*

$$\mathbb{P}\left(t^\lambda < \infty \mid p_0\right) = \left(\frac{\lambda}{p_0}\right)^{-1/2} \exp\left(-\frac{1}{2}|\log(\lambda/p_0)|\right) = 1 \wedge (\frac{p_0}{\lambda}). \tag{3.9}$$

3.3.2 Application

Equipped with the results in Section 3.3.1, we can compute the value of an early exit strategy V^λ defined in (3.3) as following:

$$\begin{aligned} V^\lambda(p_0) &= \mathbb{E}\left[g_R(p_{t^\lambda}) + \frac{1}{2}\int_0^{t^\lambda} \sigma^2 p_s^{1/2} \mathbf{1}_{p_s \in R} \mathrm{d}s \mid p_0\right] \\ &= \mathbb{E}\left[g_R(p_{t^\lambda})\right] + \underbrace{\frac{1}{2}\int_{p_\ell}^{p_u} \frac{\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right]}{b^{3/2}} \mathrm{d}b}_{=:F^\lambda(p_0)}, \end{aligned} \tag{3.10}$$

where at the second equality, we have applied the occupation time formula and the Fubini theorem. Splitting the expectation according to $\{t^\lambda < \infty\}$ or $\{t^\lambda = \infty\}$, and using (3.9), we get that the first term in (3.10) equals

$$\mathbb{E}\left[g_R(p_{t^\lambda})\right] = g_R(\lambda)\left(1 \wedge (\frac{p_0}{\lambda})\right),$$

since $g_R(0) = 0$. For the second term, we can write it as

$$\int_{p_\ell}^{p_u} \frac{\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right]}{b^{3/2}} \mathrm{d}b = \int_{p_\ell}^{p_u} \frac{\mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda<\infty} \mid p_0\right]}{b^{3/2}} \mathrm{d}b + \int_{p_\ell}^{p_u} \frac{\mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda=\infty} \mid p_0\right]}{b^{3/2}} \mathrm{d}b.$$

Each expectation above can be computed as minus the derivative of the moment generating function formulas (3.7) and (3.8) computed at $\gamma = 0$. Applying the procedure, we get the following result:

$$\mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda<\infty} \mid p_0\right] \tag{3.11}$$

$$= b^2\sigma^2 \times \frac{2}{b\sigma^2}(1 \wedge (\frac{p_0}{\lambda})) \times \begin{cases} 0 & \text{in case (i): } p_0 \le \lambda \le b, \quad b \le \lambda \le p_0, \\ (\mathrm{e}^{-|\log(p_0/b)|} - \mathrm{e}^{-|\log(\lambda/b)|}), & \text{in case (ii): } b \wedge \lambda \le p_0 \le b \vee \lambda, \\ (1 - \mathrm{e}^{-|\log(\lambda/b)|}), & \text{in case (iii): } \lambda \le b \le p_0, \quad p_0 \le b \le \lambda. \end{cases}$$

For $\lambda > b$, we have

$$\begin{aligned} &\mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda=\infty} \mid p_0\right] \\ &= b^2\sigma^2 \times \frac{2}{b\sigma^2} \times \\ &\begin{cases} 0 & \text{in case (iv): } p_0 \le \lambda \le b, \quad b \le \lambda \le p_0, \\ (1 - \mathrm{e}^{-|\log(p_0/\lambda)|})(1 - \mathrm{e}^{-|\log(\lambda/b)|}), & \text{in case (v): } b \wedge \lambda \le p_0 \le b \vee \lambda, \\ (\mathrm{e}^{-|\log(p_0/b)|} - \mathrm{e}^{-|\log(p_0/\lambda)|})(1 - \mathrm{e}^{-|\log(\lambda/b)|}), & \text{in case (vi): } \lambda \le b \le p_0, \quad p_0 \le b \le \lambda. \end{cases} \end{aligned} \tag{3.12}$$

This gives us the following result for the expected swap fees.

Proposition 3.8. *For a geometric Brownian motion price process, $p_t = p_0 \exp(-\frac{1}{2}\sigma^2 t + \sigma W_t)$, the expected swap fees $F^\lambda(p_0) := \frac{1}{2}\int_{p_\ell}^{p_u} \frac{\mathbb{E}\left[A^b_{t^\lambda}(p)|p_0\right]}{b^{3/2}}\mathrm{d}b$ is given as*

- *case $\lambda \le p_0$:*

$$F^\lambda(p_0) = \begin{cases} 2(p_0 - \lambda)(p_\ell^{-1/2} - p_u^{-1/2}), & \lambda \le p_0 \le p_\ell, \\ 2(\sqrt{p_0} - \sqrt{\lambda}) + 2\lambda(p_0^{-1/2} - \lambda^{-1/2}) + 2(p_0 - \lambda)(p_0^{-1/2} - p_u^{-1/2}), & p_\ell \le \lambda \le p_0 \le p_u, \\ 0 & p_u \le \lambda \le p_0, \\ 2(\sqrt{p_0} - \sqrt{p_\ell}) + 2\lambda(p_0^{-1/2} - p_\ell^{-1/2}) + 2(p_0 - \lambda)(p_0^{-1/2} - p_u^{-1/2}), & \lambda \le p_\ell \le p_0 \le p_u, \\ 2(\sqrt{p_u} - \sqrt{p_\ell}) + 2\lambda(p_u^{-1/2} - p_\ell^{-1/2}), & \lambda \le p_\ell \le p_u \le p_0, \\ 2(\sqrt{p_u} - \sqrt{\lambda}) + 2\lambda(p_u^{-1/2} - \lambda^{-1/2}) & p_\ell \le \lambda \le p_u \le p_0. \end{cases}$$

- *case $\lambda \ge p_0$:*

$$F^\lambda(p_0) = \begin{cases} 0, & p_0 \le \lambda \le p_\ell, \\ 2\left(1 - \frac{p_0}{\lambda}\right)(\sqrt{p_0} - \sqrt{p_\ell}) + 2p_0(p_0^{-1/2} - \lambda^{-1/2}) - \frac{2p_0}{\lambda}(\sqrt{\lambda} - \sqrt{p_0}), & p_\ell \le p_0 \le \lambda \le p_u, \\ 2\left(1 - \frac{p_0}{\lambda}\right)(\sqrt{p_u} - \sqrt{p_\ell}), & p_u \le p_0 \le \lambda, \\ 2p_0(p_\ell^{-1/2} - \lambda^{-1/2}) - \frac{2p_0}{\lambda}(\sqrt{\lambda} - \sqrt{p_\ell}), & p_0 \le p_\ell \le \lambda \le p_u, \\ 2p_0(p_\ell^{-1/2} - p_u^{-1/2}) - \frac{2p_0}{\lambda}(\sqrt{p_u} - \sqrt{p_\ell}), & p_0 \le p_\ell \le p_u \le \lambda, \\ 2(1 - \frac{p_0}{\lambda})(\sqrt{p_0} - \sqrt{p_\ell}) + 2p_0(p_0^{-1/2} - p_u^{-1/2}) - \frac{2p_0}{\lambda}(\sqrt{p_u} - \sqrt{p_0}), & p_\ell \le p_0 \le p_u \le \lambda. \end{cases}$$

We provide detailed calculations in Section A.2 of the appendix. It is thus clear that $V^\lambda(p_0)$ (3.10) is independent of σ. This is coherent with the time-change arguments used in the proof of Theorem 3.4. From the formulas obtained in Proposition 3.8, we also retrieve that the optimal λ is given by $\lambda = 0$. See Figure 3 where we plot V^λ for λ varying between 0 and $5p_u$ when $R = [p_\ell, p_u]$. This is coherent with the result of Theorem 3.3, which states that LP is better off to stay in the liquidity pool forever. The advantage of using an early-exit strategy associated to t^λ is to get an expected reward, which is close to the optimal one, without waiting forever.

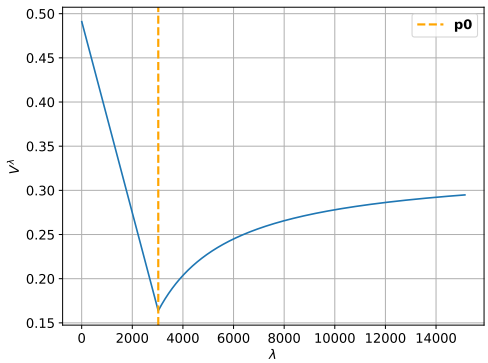

Figure 3: V^λ as a function of λ.

4 Investment return on minting liquidity

4.1 Problem formulation

As it is clear from our results and discussion from the previous section, in the case when the swap fees and pool holding value are not discounted, the optimal stopping level corresponds to an infinite stopping time ("stay in the pool forever"). To better measure the benefit for an LP in doing so, we introduce the concept of an equivalent rate of return for an LP who mints liquidity on a unitary range. Essentially, we compare the return of an LP to a fixed return on a static portfolio holding by applying a nonzero discounting rate $c > 0$ to both the pool holding value and swap fees. By using standard arguments, we can define a new value function equivalent to (3.1) and (3.10) in the setting of nonzero discounting factor as following

$$\begin{aligned} V^{\lambda,c}(p_0) &:= \mathbb{E}\left[\mathrm{e}^{-ct^\lambda} g_R(p_{t^\lambda}) + p_{t^\lambda}\frac{\sigma^2}{4}\int_0^{t^\lambda} \mathrm{e}^{-cs} p_s^{-1/2}\mathbf{1}_{p_s\in R}\mathrm{d}s + \frac{\sigma^2}{4}\int_0^{t^\lambda} \mathrm{e}^{-cs} p_s^{1/2}\mathbf{1}_{p_s\in R}\mathrm{d}s \mid p_0\right] \\ &= \mathbb{E}\left[\mathrm{e}^{-ct^\lambda} g_R(p_{t^\lambda}) + \frac{\sigma^2}{2}\int_0^{t^\lambda} \mathrm{e}^{-cs} p_s^{1/2}\mathbf{1}_{p_s\in R}\mathrm{d}s \mid p_0\right]. \end{aligned} \tag{4.1}$$

If the best exit level is chosen by the LP, then the best value is given as

$$V^{\star,c}(p_0) := \sup_{\lambda>0} V^{\lambda,c}(p_0).$$

If c is too high, one could expect that the above value is equal to the initial investment $g_R(p_0)$. In other words, the LP can not beat such a high target return and is better off to exit immediately, that is, $\lambda = p_0$. This heuristic principle allows us to introduce the concept of equivalent interest rate of return $c^\star$. It is defined as the maximal discounting factor used to compute $V^{\lambda,c}(p_0)$ such that there is indifference between keeping the liquidity and burning it immediately. We set

$$c^\star := \max\left\{c : \frac{V^{\star,c}(p_0)}{g_R(p_0)} > 1\right\}. \tag{4.2}$$

Note that $c^\star$ a priori depends on p_0, on the unitary range R and on the volatility σ. Determining its value is helpful for an LP to decide on which range they should deposit their liquidity.

4.2 Numerical computation

In (4.1), it is not possible to compute a closed-form expression for $V^{\lambda,c}(p_0)$ and we resort to numerically solving the associated differential equation for it. For this purpose, we first decompose $V^{\lambda,c}(p_0)$ into two pieces:

$$V^{\lambda,c}(p_0) := V_1^{\lambda,c}(p_0) + \frac{1}{2}\widetilde{V}^{\lambda,c}(p_0) \qquad \text{with} \qquad \begin{cases} V_1^{\lambda,c}(p_0) := \mathbb{E}\left[\mathrm{e}^{-ct^\lambda} g_R(p_{t^\lambda}) \mid p_0\right], \\ \widetilde{V}^{\lambda,c}(p_0) := \mathbb{E}\left[\sigma^2 \int_0^{t^\lambda} \mathrm{e}^{-cs} p_s^{1/2} \mathbf{1}_{p_s\in R} \mathrm{d}s \mid p_0\right]. \end{cases}$$

Term $V_1^{\lambda,c}(p_0)$. We have

$$V_1^{\lambda,c}(p_0) = g_R(\lambda)\mathbb{E}\left[\mathrm{e}^{-ct^\lambda}\mathbf{1}_{t^\lambda<\infty}|p_0\right]. \tag{4.3}$$

From [BS15, Formula 2.0.1, Page 622], we have

$$\mathbb{E}\left[\mathrm{e}^{-ct^\lambda}\mathbf{1}_{t^\lambda<\infty}|p_0\right] = \begin{cases} \left(\frac{p_0}{\lambda}\right)^{\sqrt{\frac{1}{4}+\frac{2c}{\sigma^2}}+\frac{1}{2}}, & p_0 \le \lambda, \\ \left(\frac{\lambda}{p_0}\right)^{\sqrt{\frac{1}{4}+\frac{2c}{\sigma^2}}-\frac{1}{2}}, & p_0 \ge \lambda. \end{cases} \tag{4.4}$$

Term $\widetilde{V}^{\lambda,c}(p_0)$. This term we solve numerically. By the Feynman-Kac formula, we get the following Dirichlet problem satisfied by $\widetilde{V}^{\lambda,c}$

$$\begin{cases} \frac{1}{2}\sigma^2 x^2 \widetilde{V}_{xx}^{\lambda,c} - c\widetilde{V}^{\lambda,c} + \sigma^2 x^{1/2}\mathbf{1}_{x\in R} = 0, & x \in D, \\ \widetilde{V}^{\lambda,c}(\lambda) = 0. \end{cases} \tag{4.5}$$

In the above, the domain D is given as

$$D = \begin{cases} (\lambda, \infty), & p_0 > \lambda, \\ (0, \lambda), & p_0 < \lambda. \end{cases}$$

Recall that $R = [p_\ell, p_u]$. By a change of variable $y = \log x$, and defining $v(y) := \widetilde{V}^{\lambda,c}(x)$, we get $x\widetilde{V}_x^{\lambda,c} = v_y$ and $x^2\widetilde{V}_{xx}^{\lambda,c} = v_{yy} - v_y$. Thus, the reformulated Dirichlet problem is written as

$$\begin{cases} \frac{1}{2}(v_{yy} - v_y) - \frac{c}{\sigma^2}v + e^{y/2}\mathbf{1}_{y\in\tilde{R}} = 0, & y \in \widetilde{D}, \\ v(\log\lambda) = 0, \end{cases} \tag{4.6}$$

where the modified domain $\widetilde{D}$ is given as

$$\widetilde{D} = \begin{cases} (\log\lambda, \infty), & p_0 > \lambda, \\ (-\infty, \log\lambda), & p_0 < \lambda. \end{cases}$$

In the above $\tilde{R} = [\log p_\ell, \log p_u] =: [\tilde{p}_\ell, \tilde{p}_u]$. It is clear from the ordinary differential equation (ODE) (4.6), that the solution depends on $\frac{c}{\sigma^2}$ for a given λ and a given range R. To compute $c^\star$ defined in (4.2), we solve (4.6) numerically using the finite difference method. We provide complete implementation details in Section B.

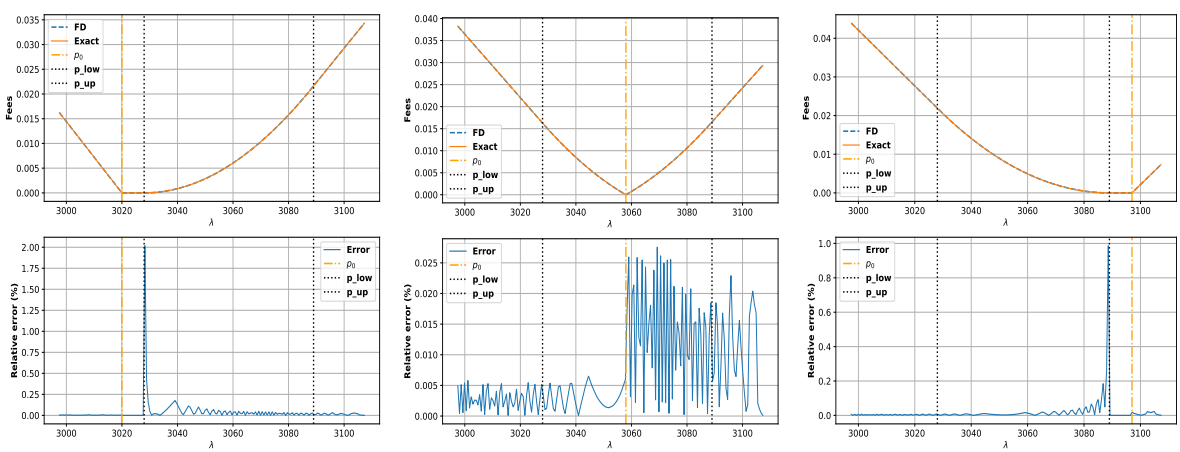

Figure 4: $V^{\lambda,c}(p_0)$ comparison using the finite difference (FD) approximation and the exact local time formula for different values of p_0 and $c = 0$. On the left (resp. middle, resp. right), p_0 is below (resp. inside, resp. above) the range R.

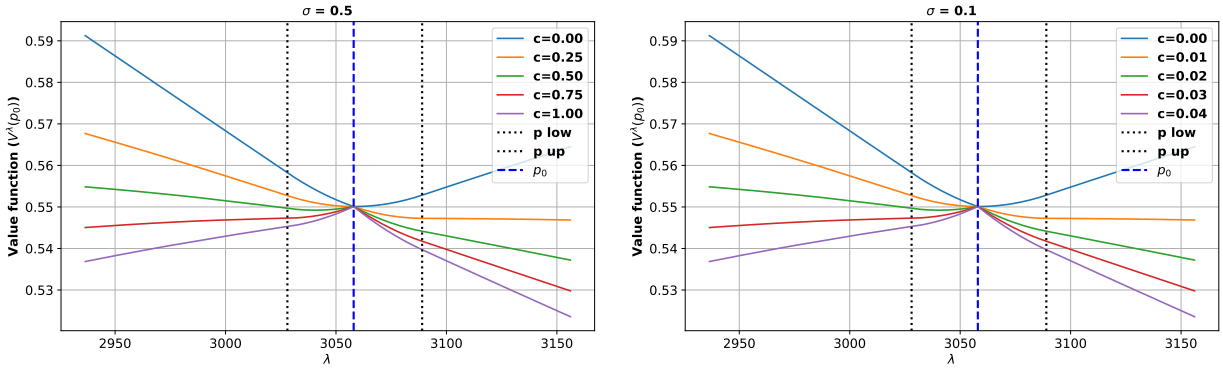

Figure 5: Value function $V^{\lambda,c}(p_0)$ for different values of c.

Results. We first test the accuracy of our finite difference method by comparing it with the closed-form formula for $V^\lambda(p_0)$ derived in Section 3.3.2 for $c = 0$. For tick range $[80160, 80360]$, we get the unitary price range $[p_\ell, p_u]$ roughly as $[3028, 3089]$. Moreover, to approximate the domain D, we set $p_{small} = 4$, $p_{large} = 5p_u$, and the number of finite difference steps $N = 1 \times 10^6$. The accuracy comparison is presented in Figure 4 for three different values of $p_0 = 3020, 3058, 3097$. It is clear that for large enough N, our finite difference method is quite accurate.
Next, for two different values of σ, we illustrate the effect of c on $V^{\lambda,c}(p_0)$ for a chosen value of p_0 and R. In Figure 5, for the price range $[3028, 3089]$, $p_0 = 3058$, we plot $V^{\lambda,c}(p_0)$ with equivalent levels of $\frac{c}{\sigma^2}$ and demonstrate that the ratio plays a deciding factor in the behaviour of $V^{\lambda,c}(p_0)$. This is also expected from the ODE (4.6).
To obtain the expected rate of return defined in (4.2), we then set $\sigma = 1$ and then numerically estimate $c^\star$ for different values of p_0 in the range $[3028, 3089]$. To obtain $c^\star$ for a given p_0, we compare the value of $V^{\lambda,c}(p_0)$ at $\lambda \in \{0, p_0, \infty\}$. The results are shown in Figure 6. We can conclude that minting liquidity on the current range is the most profitable strategy for the LP and the best possible $c^\star$ on the price range $[3028, 3089]$ is 42.5%. But on the other hand, the LP may have to stay in the pool forever to obtain such high values of $c^\star$.

As explained before, $V^{\lambda,c}(p_0)$ depends on c and σ through the ratio c/σ^2: see (4.3)-(4.4) for the term $V_1^{\lambda,c}$, and see (4.5) for the term $\widetilde{V}^{\lambda,c}$. Hence, for a volatility $\sigma > 0$, the best interest rate of return is

$$0.425 \cdot \sigma^2.$$

To exemplify this formula, consider a common value of volatility like 50%. In this case, the best expected return is about 10.6% when the LP invests in the range containing the current price p_0.

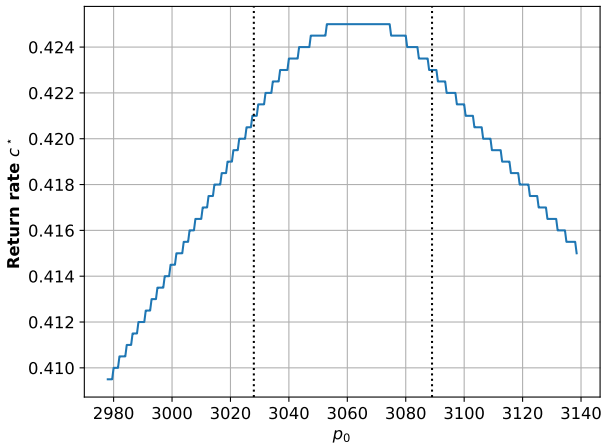

Figure 6: Interest rate of return $c^\star$ for different values of p_0.

5 Conclusion

This work enhances the understanding of Uniswap v3's dynamics, providing liquidity providers with actionable insights to optimize their strategies. In particular, our work has focused on the problem of optimal stopping for liquidity exit. By framing the issue within the context of optimal stopping theory, we have derived significant findings: the importance of strategically liquidating each price range rather than opting for a simultaneous exit, and the result that under certain conditions, remaining in the liquidity pool indefinitely can be optimal. Furthermore, our analysis introduces an equivalent rate of return for LPs and establishes that under a Black-Scholes framework, the best return depends on the volatility of the assets involved, following the relationship $0.425\sigma^2$. This return is achieved by investing directly in the range where the current price is located.

A Proofs

A.1 Proof of Proposition 2.1

Denote by $(\rho(i))_{i\geq 0}$ where $\rho(i) := [\pi_{l_i}, \pi_{l_{i+1}})$ the sequence of unitary square root price changes in the pool with $\pi_{l_i} := \sqrt{p(i\delta_\pi)}$. Recall the relationship between price p and tick in (2.1). Furthermore, denote by $R(i) := [p(i\delta_\pi), p((i+1)\delta_\pi))$. The objective function at any time t for a liquidity curve $(\Delta L_\pi)_\pi$ in (2.6) can then be written as follows:

$$J\big(t, (\Delta L_\pi)_\pi\big) = \sum_{i=-\infty}^{+\infty} \Delta L_{\pi_{l_i}} J_{R(i)}(t),$$

where

$$J_{R(i)}(t) := p_t \int_{\pi_t}^{\infty} \mathbf{1}_{\pi\in\rho(i)} \frac{1}{\pi^2} \mathrm{d}\pi + \int_0^{\pi_t} \mathbf{1}_{\pi\in\rho(i)} \mathrm{d}\pi + p_t \int_{R(i)} \frac{A_t^b(p)}{4b^{5/2}} \mathrm{d}b + \int_{R(i)} \frac{A_t^b(p)}{4b^{3/2}} \mathrm{d}b.$$

Note that the sum over i is finite since an LP mints liquidity on a finite number of price ranges $R(i)$. Thus, we have a linearity property in liquidity for the objective function. Now, as the supremum is sub-linear, we have

$$\sup_{t\in\mathcal{T}(\mathbb{R}^+)} \mathbb{E}\left[J\big(t,(\Delta L_\pi)_\pi\big]\right) = \sup_{t\in\mathcal{T}(\mathbb{R}^+)} \sum_{i=-\infty}^{+\infty} \Delta L_{\pi_{l_i}} \mathbb{E}\left[J_{R(i)}(t)\right] \leq \sum_{i=-\infty}^{+\infty} \Delta L_{\pi_{l_i}} \sup_{t\in\mathcal{T}(\mathbb{R}^+)} \mathbb{E}\left[J_{R(i)}(t)\right].$$

□

A.2 Proof of Proposition 3.8

Overall, we have the following permutations of the starting price p_0 and threshold λ with respect to a fixed price interval R with lower bound p_ℓ and upper bound p_u :

Case Index	Situation	Relevance of contribution $\{t^\lambda < \infty\}$	Relevance of contribution $\{t^\lambda = \infty\}$
1	$\lambda \leq p_0 \leq p_\ell \leq p_u$	✓	✗
2	$p_0 \leq \lambda \leq p_\ell \leq p_u$	✗	✗
3	$p_\ell \leq \lambda \leq p_0 \leq p_u$	✓	✗
4	$p_\ell \leq p_0 \leq \lambda \leq p_u$	✓	✓
5	$p_\ell \leq p_u \leq \lambda \leq p_0$	✗	✗
6	$p_\ell \leq p_u \leq p_0 \leq \lambda$	✓	✓
7	$\lambda \leq p_\ell \leq p_0 \leq p_u$	✓	✗
8	$\lambda \leq p_\ell \leq p_u \leq p_0$	✓	✗
9	$p_0 \leq p_\ell \leq \lambda \leq p_u$	✓	✓
10	$p_\ell \leq \lambda \leq p_u \leq p_0$	✓	✗
11	$p_0 \leq p_\ell \leq p_u \leq \lambda$	✓	✓
12	$p_\ell \leq p_0 \leq p_u \leq \lambda$	✓	✓

For each of the relevant situations, we provide detailed computation of $\int_{p_\ell}^{p_u} \frac{\mathbb{E}\left[A_{t^\lambda}^b(p)\mathbf{1}_{t^\lambda<\infty}|p_0\right]}{b^{3/2}} \mathrm{d}b$ and $\int_{p_\ell}^{p_u} \frac{\mathbb{E}\left[A_{t^\lambda}^b(p)\mathbf{1}_{t^\lambda=\infty}|p_0\right]}{b^{3/2}} \mathrm{d}b$.

Case 1) $\lambda \leq p_0 \leq p_\ell \leq b \leq p_u$: In this case, using (3.11)-case (ii), we get

$$\mathbb{E}\left[A_{t^\lambda}^b(p) \mid p_0\right] = \mathbb{E}\left[A_{t^\lambda}^b(p)\mathbf{1}_{t^\lambda<\infty} \mid p_0\right]$$

$$= b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{(\mathrm{e}^{-|\log(b/p_0)|} - \mathrm{e}^{-|\log(b/\lambda)|})}{\frac{1}{2}b\sigma^2}$$
$$= b^2\sigma^2 \times \frac{2(p_0-\lambda)}{b^2\sigma^2} = 2(p_0-\lambda).$$

Then,

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b = 4(p_0-\lambda)(p_\ell^{-1/2} - p_u^{-1/2}).$$

Case 2) $p_0 \le \lambda \le p_\ell \le p_u$: This case leads to no contribution.

Case 3) $p_\ell \le \lambda \le p_0 \le p_u$: In this case, we will burn the liquidity when price hits the level $\lambda \ge p_\ell$ starting from $p_0 \ge p_\ell$. Thus, we can write the following

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b = \int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda<\infty} \mid p_0\right] \mathrm{d}b$$
$$= \int_{\lambda}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda<\infty} \mid p_0\right] \mathrm{d}b.$$

Now, based on the formula in Equation (3.11), we have to consider two different situations: $b \in [\lambda, p_0]$ (case (iii)) and $b \in [p_0, p_u]$ (case (ii)). For case (iii), we get

$$\mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda<\infty} \mid p_0\right] = b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{2(1-\mathrm{e}^{-|\log(\lambda/b)|})}{b\sigma^2}$$
$$= b^2\sigma^2 \times \frac{2(b-\lambda)}{b^2\sigma^2} = 2(b-\lambda).$$

For case (ii), we get

$$\mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda<\infty} \mid p_0\right] = b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{2(\mathrm{e}^{-|\log(b/p_0)|} - \mathrm{e}^{-|\log(b/\lambda)|})}{b\sigma^2}$$
$$= b^2\sigma^2 \times \frac{2(p_0-\lambda)}{b^2\sigma^2} = 2(p_0-\lambda).$$

All in all, we get

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p)\mathbf{1}_{t^\lambda<\infty} \mid p_0\right] \mathrm{d}b$$
$$= \int_{\lambda}^{p_0} \frac{1}{b^{3/2}} 2(b-\lambda)\mathrm{d}b + \int_{p_0}^{p_u} \frac{1}{b^{3/2}} 2(p_0-\lambda)\mathrm{d}b$$
$$= 4(\sqrt{p_0} - \sqrt{\lambda}) + 4\lambda(p_0^{-1/2} - \lambda^{-1/2}) + 4(p_0-\lambda)(p_0^{-1/2} - p_u^{-1/2}).$$

Case 4) $p_\ell \le p_0 \le \lambda \le p_u$: In this case, we will burn the liquidity when price hits the level $\lambda \le p_u$ starting from $p_0 \le p_u$. Thus, we can write the following

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b = \int_{p_\ell}^{\lambda} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b.$$

Based on the formulas in Equations (3.11)-(3.12), we have to consider two different situations: $b \in [p_\ell, p_0]$ (cases (ii) and (v)) and $b \in [p_0, \lambda]$ (cases (iii) and (vi)). For $b \in [p_\ell, p_0]$, we get

$$\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{2(\mathrm{e}^{-|\log(b/p_0)|} - \mathrm{e}^{-|\log(b/\lambda)|})}{b\sigma^2} \\
&\quad + b^2\sigma^2 \times \frac{2(1-\mathrm{e}^{-|\log(p_0/\lambda)|})(1-\mathrm{e}^{-|\log(\lambda/b)|})}{b\sigma^2} \\
&= b^2\sigma^2 \times \frac{p_0}{\lambda}\frac{2}{b\sigma^2}\left(\frac{b}{p_0} - \frac{b}{\lambda}\right) + b^2\sigma^2 \times \frac{2}{b\sigma^2}\left(1 - \frac{p_0}{\lambda}\right)\left(1 - \frac{b}{\lambda}\right) \\
&= b^2\sigma^2 \times \frac{2}{b\sigma^2}\left(1 - \frac{p_0}{\lambda}\right) = 2b\left(1 - \frac{p_0}{\lambda}\right).
\end{aligned}$$

For $b \in [p_0, \lambda]$, we get

$$\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{2(1-\mathrm{e}^{-|\log(\lambda/b)|})}{b\sigma^2} \\
&\quad + b^2\sigma^2 \times \frac{2}{b\sigma^2}(\mathrm{e}^{-|\log(p_0/b)|} - \mathrm{e}^{-|\log(p_0/\lambda)|})(1-\mathrm{e}^{-|\log(\lambda/b)|}) \\
&= b^2\sigma^2 \times \frac{p_0}{\lambda}\frac{2}{b\sigma^2}\left(1 - \frac{b}{\lambda}\right) + b^2\sigma^2 \times \frac{2}{b\sigma^2}\left(\frac{p_0}{b} - \frac{p_0}{\lambda}\right)\left(1 - \frac{b}{\lambda}\right) \\
&= b^2\sigma^2 \times \frac{2p_0}{b^2\sigma^2}\left(1 - \frac{b}{\lambda}\right) = 2p_0\left(1 - \frac{b}{\lambda}\right).
\end{aligned}$$

Combining all, we derive

$$\begin{aligned}
&\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b \\
&= \int_{p_\ell}^{p_0} \frac{2}{b^{1/2}}\left(1 - \frac{p_0}{\lambda}\right) \mathrm{d}b + \int_{p_0}^{\lambda} \frac{2p_0}{b^{3/2}}\left(1 - \frac{b}{\lambda}\right) \mathrm{d}b \\
&= 4\left(1 - \frac{p_0}{\lambda}\right)(\sqrt{p_0} - \sqrt{p_\ell}) + 4p_0(p_0^{-1/2} - \lambda^{-1/2}) - \frac{4p_0}{\lambda}(\sqrt{\lambda} - \sqrt{p_0}).
\end{aligned}$$

Case 5) $p_\ell \le p_u \le \lambda \le p_0$: This case leads to no contribution.

Case 6) $p_\ell \le b \le p_u \le p_0 \le \lambda$: In this case, we burn the liquidity when we price hits the level $\lambda \ge p_0$. Thus, we are in the situation of cases (ii) and (v) of Equations (3.11)-(3.12). We can get the following

$$\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{2(\mathrm{e}^{-|\log(b/p_0)|} - \mathrm{e}^{-|\log(b/\lambda)|})}{b\sigma^2} \\
&\quad + b^2\sigma^2 \times \frac{2(1-\mathrm{e}^{-|\log(p_0/\lambda)|})(1-\mathrm{e}^{-|\log(\lambda/b)|})}{b\sigma^2} \\
&= b^2\sigma^2 \times \frac{p_0}{\lambda}\frac{2}{b\sigma^2}\left(\frac{b}{p_0} - \frac{b}{\lambda}\right) + b^2\sigma^2 \times \frac{2}{b\sigma^2}\left(1 - \frac{p_0}{\lambda}\right)\left(1 - \frac{b}{\lambda}\right) \\
&= b^2\sigma^2 \times \frac{2}{b\sigma^2}\left(1 - \frac{p_0}{\lambda}\right) = 2b\left(1 - \frac{p_0}{\lambda}\right),
\end{aligned}$$

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}}\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b = 4\left(1 - \frac{p_0}{\lambda}\right)(\sqrt{p_u} - \sqrt{p_\ell}).$$

Case 7) $\lambda \le p_\ell \le p_0 \le p_u$: In this case, we burn the liquidity when price hits the level $\lambda \le p_\ell$. Based on the Equation (3.11), we have to consider two different situations: $b \in [p_\ell, p_0]$ (case (iii)) and $b \in [p_0, p_u]$ (case (ii)). For case (iii), we get

$$\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{2(1-\mathrm{e}^{-|\log(b/\lambda)|})}{b\sigma^2} \\
&= b^2\sigma^2 \times \frac{2}{b\sigma^2}\left(1 - \frac{\lambda}{b}\right) = 2(b-\lambda).
\end{aligned}$$

For case (ii), we get

$$\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{2(\mathrm{e}^{-|\log(b/p_0)|} - \mathrm{e}^{-|\log(b/\lambda)|})}{b\sigma^2} \\
&= b^2\sigma^2 \times \frac{2}{b\sigma^2}\left(\frac{p_0}{b} - \frac{\lambda}{b}\right) = b^2\sigma^2 \times \frac{2(p_0-\lambda)}{\sigma^2 b^2} = 2(p_0-\lambda).
\end{aligned}$$

It directly follows that

$$\begin{aligned}
&\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}}\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b \\
&= \int_{p_\ell}^{p_0} \frac{2(b-\lambda)}{b^{3/2}}\mathrm{d}b + \int_{p_0}^{p_u} \frac{2(p_0-\lambda)}{b^{3/2}}\mathrm{d}b \\
&= 4(\sqrt{p_0} - \sqrt{p_\ell}) + 4\lambda(p_0^{-1/2} - p_\ell^{-1/2}) + 4(p_0-\lambda)(p_0^{-1/2} - p_u^{-1/2}).
\end{aligned}$$

Case 8) $\lambda \le p_\ell \le b \le p_u \le p_0$: In this case, use Equation (3.11) – case (iii) to get

$$\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] = b^2\sigma^2 \times (1 \wedge (\frac{p_0}{\lambda})) \times \frac{2(1-\mathrm{e}^{-|\log(b/\lambda)|})}{b\sigma^2}$$

$$= b^2\sigma^2 \times \frac{2(b-\lambda)}{b^2\sigma^2} = 2(b-\lambda),$$

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b = 4(\sqrt{p_u} - \sqrt{p_\ell}) + 4\lambda(p_u^{-1/2} - p_\ell^{-1/2}).$$

Case 9) $p_0 \le p_\ell \le \lambda \le p_u$: In this case, we burn the liquidity when price increases and moves up to a level $\lambda \le p_u$. Thus, we have the following

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b = \int_{p_\ell}^{\lambda} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b.$$

Based on Equations (3.11) and (3.12), we are in the situation of cases (iii) and (vi):

$$\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times \frac{2}{b\sigma^2}(1 \wedge (\frac{p_0}{\lambda}))(1 - \mathrm{e}^{-|\log(\lambda/b)|}) \\
&\quad + b^2\sigma^2 \times \frac{2}{b\sigma^2}(\mathrm{e}^{-|\log(p_0/b)|} - \mathrm{e}^{-|\log(p_0/\lambda)|})(1 - \mathrm{e}^{-|\log(\lambda/b)|}) \\
&= b^2\sigma^2 \times \frac{p_0}{\lambda}\frac{2}{b\sigma^2}\Big(1 - \frac{b}{\lambda}\Big) + b^2\sigma^2 \times \frac{2}{b\sigma^2}(\frac{p_0}{b} - \frac{p_0}{\lambda})\Big(1 - \frac{b}{\lambda}\Big) \\
&= b^2\sigma^2 \times \frac{2p_0}{b^2\sigma^2}\Big(1 - \frac{b}{\lambda}\Big) = 2p_0\Big(1 - \frac{b}{\lambda}\Big),
\end{aligned}$$

$$\begin{aligned}
\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b &= \int_{p_\ell}^{\lambda} \frac{2p_0}{b^{3/2}}\Big(1 - \frac{b}{\lambda}\Big)\mathrm{d}b \\
&= 4p_0(p_\ell^{-1/2} - \lambda^{-1/2}) - \frac{4p_0}{\lambda}(\sqrt{\lambda} - \sqrt{p_\ell}).
\end{aligned}$$

Case 10) $p_\ell \le \lambda \le p_u \le p_0$: In this case, we burn the liquidity when price decreases to a level $p_\ell \le \lambda \le p_u$. Thus, we have the following:

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b = \int_{\lambda}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b.$$

Invoking Equation (3.11) – case (iii), we readily obtain:

$$\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times \Big(1 \wedge (\frac{p_0}{\lambda})\Big) \times \frac{2(1 - \mathrm{e}^{-|\log(b/\lambda)|})}{b\sigma^2} \\
&= b^2\sigma^2 \times \frac{2}{b\sigma^2}\Big(1 - \frac{\lambda}{b}\Big) = 2(b-\lambda),
\end{aligned}$$

$$\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}} \mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b = 4(\sqrt{p_u} - \sqrt{\lambda}) + 4\lambda(p_u^{-1/2} - \lambda^{-1/2}).$$

Case 11) $p_0 \le p_\ell \le p_u \le \lambda$: In this case, we burn the liquidity when price increases to a level $\lambda \ge p_u$. Using Equations (3.11) and (3.12) in cases (iii) and (vi) respectively, we derive

$$\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] = b^2\sigma^2 \times \Big(1 \wedge (\frac{p_0}{\lambda})\Big) \frac{2(1 - \mathrm{e}^{-|\log(\lambda/b)|})}{b\sigma^2}$$

$$
\begin{aligned}
&+ b^2\sigma^2 \times \frac{2}{b\sigma^2}(\mathrm{e}^{-|\log(p_0/b)|} - \mathrm{e}^{-|\log(p_0/\lambda)|})(1 - \mathrm{e}^{-|\log(\lambda/b)|}) \\
&= b^2\sigma^2 \times \frac{p_0}{\lambda}\frac{2}{b\sigma^2}\Big(1 - \frac{b}{\lambda}\Big) + \frac{2}{b\sigma^2}(\frac{p_0}{b} - \frac{p_0}{\lambda})(1 - \frac{b}{\lambda}) \\
&= b^2\sigma^2 \times \frac{2p_0}{b^2\sigma^2}\Big(1 - \frac{b}{\lambda}\Big) = 2p_0\Big(1 - \frac{b}{\lambda}\Big),
\end{aligned}
$$

$$
\begin{aligned}
\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}}\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b &= \int_{p_\ell}^{p_u} \frac{2p_0}{b^{3/2}}\Big(1 - \frac{b}{\lambda}\Big)\mathrm{d}b \\
&= 4p_0(p_\ell^{-1/2} - p_u^{-1/2}) - \frac{4p_0}{\lambda}(\sqrt{p_u} - \sqrt{p_\ell}).
\end{aligned}
$$

Case 12) $p_\ell \le p_0 \le p_u \le \lambda$: In view of Equations (3.11) and (3.12), we have to consider two different situations: $b \in [p_\ell, p_0]$ (cases (ii) and (v)) and $b \in [p_0, p_u]$ (cases (iii) and (vi)). When $b \in [p_\ell, p_0]$, we get

$$
\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times \frac{2}{b\sigma^2}(1 \wedge (\frac{p_0}{\lambda}))(\mathrm{e}^{-|\log(b/p_0)|} - \mathrm{e}^{-|\log(b/\lambda)|}) \\
&\quad + b^2\sigma^2 \times \frac{2}{b\sigma^2}(1 - \mathrm{e}^{-|\log(p_0/\lambda)|})(1 - \mathrm{e}^{-|\log(\lambda/b)|}) \\
&= b^2\sigma^2 \times \frac{2p_0}{b\sigma^2\lambda}\Big(\frac{b}{p_0} - \frac{b}{\lambda}\Big) + b^2\sigma^2 \times \frac{2}{b\sigma^2}(1 - \frac{p_0}{\lambda})(1 - \frac{b}{\lambda}) \\
&= b^2\sigma^2 \times \frac{2}{b\sigma^2}(1 - \frac{p_0}{\lambda}) = 2b(1 - \frac{p_0}{\lambda}).
\end{aligned}
$$

For $b \in [p_0, p_u]$, we get

$$
\begin{aligned}
\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] &= b^2\sigma^2 \times \Big(1 \wedge (\frac{p_0}{\lambda})\Big)\frac{2}{b\sigma^2}(1 - \mathrm{e}^{-|\log(\lambda/b)|}) \\
&\quad + b^2\sigma^2 \times \frac{2}{b\sigma^2}(\mathrm{e}^{-|\log(p_0/b)|} - \mathrm{e}^{-|\log(p_0/\lambda)|})(1 - \mathrm{e}^{-|\log(\lambda/b)|}) \\
&= b^2\sigma^2 \times \frac{2p_0}{b\sigma^2\lambda}\Big(1 - \frac{b}{\lambda}\Big) + b^2\sigma^2 \times \frac{2}{b\sigma^2}(\frac{p_0}{b} - \frac{p_0}{\lambda})(1 - \frac{b}{\lambda}) \\
&= b^2\sigma^2 \times \frac{2p_0}{b^2\sigma^2}(1 - \frac{b}{\lambda}) = 2p_0(1 - \frac{b}{\lambda}).
\end{aligned}
$$

Then, it readily follows that

$$
\begin{aligned}
&\int_{p_\ell}^{p_u} \frac{1}{b^{3/2}}\mathbb{E}\left[A^b_{t^\lambda}(p) \mid p_0\right] \mathrm{d}b \\
&= \int_{p_\ell}^{p_0} \frac{1}{b^{3/2}}2b(1 - \frac{p_0}{\lambda})\mathrm{d}b + \int_{p_0}^{p_u} \frac{1}{b^{3/2}}2p_0(1 - \frac{b}{\lambda})\mathrm{d}b \\
&= 4(1 - \frac{p_0}{\lambda})(\sqrt{p_0} - \sqrt{p_\ell}) + 4p_0(p_0^{-1/2} - p_u^{-1/2}) - \frac{4p_0}{\lambda}(\sqrt{p_u} - \sqrt{p_0}).
\end{aligned}
$$

B Finite difference method for (4.6)

We use the Thomas algorithm for banded systems to solve the ODE (4.6) in the interior points and set the boundary conditions accordingly. The details are as below.

B.1 The case $\lambda < p_0$

Let $\tilde{p}_u = \log(p_{large})$. Divide the log-domain $[\log\lambda, \tilde{p}_u]$ into N discrete intervals. That is, $h = \frac{\tilde{p}_u - \log\lambda}{N}$. Next we denote $y_i = \log\lambda + ih$. Then, $y_0 = \log\lambda$ and $y_N = \tilde{p}_u$. At $i = N$, the value function is constant, so we have a boundary condition that $V'(p_u) = 0$. This gives $v_y(\tilde{p}_u) = 0$. Thus, we get the following linear system when we denote $v(y_i) \approx v_i$.

$$\begin{cases} v_0 = 0, \\ \frac{1}{2}\left(1 - \frac{h}{2}\right) v_{i+1} - v_i - \frac{ch^2}{\sigma^2} v_i + \frac{1}{2}\left(1 + \frac{h}{2}\right) v_{i-1} + h^2 e^{y_i/2} \mathbf{1}_{y_i \in \tilde{R}} = 0, \\ v_N - v_{N-1} = 0. \end{cases}$$

With $i = 1$, the above writes as

$$\frac{1}{2}\left(1 - \frac{h}{2}\right) v_2 - (1 + \frac{c}{\sigma^2} h^2) v_1 + h^2 e^{y_1/2} \mathbf{1}_{y_1 \in \tilde{R}} = 0.$$

With $i = N - 1$, the above writes as

$$\frac{1}{2}\left(1 - \frac{h}{2}\right) v_{N-1} - (1 + \frac{c}{\sigma^2} h^2) v_{N-1} + \frac{1}{2}\left(1 + \frac{h}{2}\right) v_{N-2} + h^2 e^{y_{N-1}/2} \mathbf{1}_{y_{N-1} \in \tilde{R}} = 0.$$

B.2 The case $p_0 > \lambda$

Let $\tilde{p}_l = \log(p_{small})$. First divide the log-domain $[\tilde{p}_l, \log\lambda]$ into N discrete intervals. That is, $h = \frac{\log\lambda - \tilde{p}_l}{N}$. Next we denote $y_i = \tilde{p}_l + ih$. Then, $y_0 = \tilde{p}_l$ and $y_N = \log\lambda$. At $i = N$, we know that the value function is zero and at y_0 it is also zero. This gives $v(\tilde{p}_l) = 0$. Thus, we solve:

$$\begin{cases} v_0 = 0, \\ \frac{1}{2}\left(1 - \frac{h}{2}\right) v_{i+1} - (1 + \frac{c}{\sigma^2} h^2) v_i + \frac{1}{2}\left(1 + \frac{h}{2}\right) v_{i-1} + h^2 e^{y_i/2} \mathbf{1}_{y_i \in \tilde{R}} = 0, \\ v_N = 0. \end{cases}$$

With $i = 1$, the above writes as

$$\frac{1}{2}\left(1 - \frac{h}{2}\right) v_2 - (1 + \frac{c}{\sigma^2} h^2) v_1 + h^2 e^{y_1/2} \mathbf{1}_{y_1 \in \tilde{R}} = 0.$$

With $i = N - 1$, the above writes as

$$-(1 + \frac{c}{\sigma^2} h^2) v_{N-1} + \frac{1}{2}\left(1 + \frac{h}{2}\right) v_{N-2} + h^2 e^{y_{N-1}/2} \mathbf{1}_{y_{N-1} \in \tilde{R}} = 0.$$

References

[AKC+21] Guillermo Angeris, Hsien-Tang Kao, Rei Chiang, Charlie Noyes, and Tarun Chitra. An analysis of Uniswap markets. Arxiv, 2021. url=https://arxiv.org/abs/1911.03380. 2.2

[AZS+21] Hayden Adams, Noah Zinsmeister, Moody Salem, River Keefer, and Dan Robinson. Uniswap v3 core. Uniswap, Tech. Rep., 2021. `https://uniswap.org/whitepaper-v3.pdf`. 1

[BS15] Andrei N. Borodin and Paavo Salminen. Handbook of Brownian motion-facts and formulae. Springer Science & Business Media, 2015. 3.3.1, 3.5, 4.2

[CZ25] Agostino Capponi and Brian Zhu. Optimal exiting for liquidity provision in constant function market makers, February 21 2025. Available at SSRN: https://ssrn.com/abstract=5148585 or http://dx.doi.org/10.2139/ssrn.5148585. 1

[DFS25] Diogo Duarte, Lucas X. Ferreira, and Yuri F. Saporito. The term structure of implied fees in liquidity pools, July 2025. Personal communication. 1

[EGM25] Mnacho Echenim, Emmanuel Gobet, and Anne-Claire Maurice. Uniswap v3: impermanent loss modeling and swap fees asymptotic analysis. SIFIN, in revision, 2025. `https://hal.science/hal-04214315/`, version 3. 1, 1, 2.1, 2.2, 2.2, 2.2, 3.2

[GM23] Emmanuel Gobet and Anastasia Melachrinos. Decentralized finance & blockchain technology. hal preprint 04131680, 2023. Tutorial at SIAM Financial Mathematics and Engineering 2023, Jun 2023, Philadelphia, United States. 1

[LHRW21] Stefan Loesch, Nate Hindman, Mark B Richardson, and Nicholas Welch. Impermanent loss in Uniswap v3. arXiv preprint arXiv:2111.09192, 2021. 1

[PS06] Goran Peskir and Albert Shiryaev. Optimal stopping and free-boundary problems. Basel: Birkhäuser, 2006. 3.2

[RY99] Daniel Revuz and Marc Yor. Continuous martingales and Brownian motion. Comprehensive Studies in Mathematics. Berlin: Springer, third edition, 1999. 2.2, 3.1, 3.2

[Tea25] Empirica Team. Token liquidity reports on Uniswap v3. Empirica, May 2025. `https://empirica.io/uniswap-liquidity-report/`. 1